WORKED SOLUTION

STEP III 1990, Question 2

Pure • Coordinate geometry

STEP III 1990, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s3-q2

The key idea

All four points lie on the single curve $t \mapsto (t^3, t^2, t)$, the *twisted cubic*. Two lines that meet lie in a common plane, so part (i) reduces to showing that O, A, B, C are never coplanar, which is a determinant that factorises like a Vandermonde determinant. In part (ii), the condition $ab = 1$ makes the scalar product and the lengths collapse into a function of the single quantity $a^2 + 1/a^2$.

Solution

First, what does "distinct" tell us? The third coordinate of the point with parameter t is t itself, so different points have different parameters. Hence a, b, c are distinct, and since O is the point with $t = 0$, none of a, b, c is zero.

Part (i)

Suppose the lines OA and BC did meet. Two lines through a common point lie in a common plane (if they happen to coincide, any plane containing them will do). So O, A, B, C would all lie in one plane through the origin, and the position vectors $\mathbf{a} = \overrightarrow{OA}$, $\mathbf{b} = \overrightarrow{OB}$, $\mathbf{c} = \overrightarrow{OC}$ would be linearly dependent. That happens exactly when

$$D = \begin{vmatrix} a^3 & a^2 & a \\ b^3 & b^2 & b \\ c^3 & c^2 & c \end{vmatrix} = 0.$$

Taking the factor a out of the first row, b out of the second and c out of the third,

$$D = abc \begin{vmatrix} a^2 & a & 1 \\ b^2 & b & 1 \\ c^2 & c & 1 \end{vmatrix}.$$

Subtract the third row from each of the first two, then expand down the last column:

$$D = abc\begin{vmatrix}(a-c)(a+c) & a-c & 0\\ (b-c)(b+c) & b-c & 0\\ c^2 & c & 1\end{vmatrix} = abc\,(a-c)(b-c)\begin{vmatrix}a+c & 1\\ b+c & 1\end{vmatrix}$$
$$= abc\,(a-b)(a-c)(b-c).$$

Every factor is non-zero: $a, b, c \neq 0$ and a, b, c are distinct. So $D \neq 0$, the four points are not coplanar, and therefore the lines OA and BC cannot intersect. (They are not parallel either, so they are skew.)

Ask yourself Why is it enough to show that the four points are not coplanar?

Because intersecting lines always span a plane. If OA and BC met at a point P, the plane through P containing both lines would contain O, A, B and C. We have shown that no such plane exists.

Part (ii)

Put $b = 1/a$. Then $a \neq 0$, and $a \neq \pm 1$, since otherwise $b = a$ and A, B would coincide. The scalar product is

$$\mathbf{a}\cdot\mathbf{b} = (ab)^3 + (ab)^2 + ab = 3,$$

and the squared lengths are

$$|\mathbf{a}|^2 = a^2(a^4+a^2+1), \qquad |\mathbf{b}|^2 = \frac{1}{a^6}+\frac{1}{a^4}+\frac{1}{a^2} = \frac{a^4+a^2+1}{a^6}.$$

Multiplying, $|\mathbf{a}|^2|\mathbf{b}|^2 = (a^4+a^2+1)^2/a^4$, so $|\mathbf{a}||\mathbf{b}| = (a^4+a^2+1)/a^2$, which is positive. Writing $\theta = \angle AOB$,

$$\cos\theta = \frac{3a^2}{a^4+a^2+1} = \frac{3}{u+1}, \qquad \text{where } u = a^2 + \frac{1}{a^2}.$$

Which values can occur. By AM–GM (or since $a^2 + a^{-2} - 2 = (a - a^{-1})^2$), $u \geq 2$, with equality only when $a^2 = 1$, which is excluded. So $u > 2$, and therefore

$$0 < \cos\theta = \frac{3}{u+1} < \frac{3}{3} = 1,$$

which gives $0 < \theta < \frac{1}{2}\pi$. No other angle is possible.

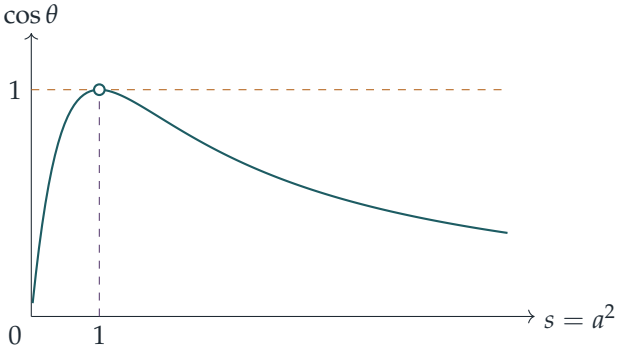
Every such value occurs. Take any θ with $0 < \theta < \frac{1}{2}\pi$, so $0 < \cos\theta < 1$. We need $u = 3/\cos\theta - 1$, which is greater than 2. The equation $a^2 + 1/a^2 = u$ is the quadratic $(a^2)^2 - u\,a^2 + 1 = 0$ in a^2, with root

$$a^2 = \frac{u + \sqrt{u^2-4}}{2} > 1,$$

which is real because $u > 2$. Choose a to be the positive square root and $b = 1/a$. Then $a > 1 > b > 0$, so $ab = 1$, $A \neq B$, neither is O, and $\cos\angle AOB = 3/(u+1) = \cos\theta$. Hence $\angle AOB = \theta$, since both angles lie in $[0, \pi]$, where cosine is one-to-one.

So $\angle AOB$ takes every value in $0 < \angle AOB < \frac{1}{2}\pi$ and no others, as required.

The graph shows $\cos\theta$ against $s = a^2$. It rises to 1 at the excluded value $s = 1$ and tends to 0 at both ends, so its values fill $(0, 1)$ exactly.



Checks

- Try $a = 2$, $b = \frac{1}{2}$: $A = (8, 4, 2)$ and $B = (\frac{1}{8}, \frac{1}{4}, \frac{1}{2})$ give $\mathbf{a} \cdot \mathbf{b} = 1 + 1 + 1 = 3$, and the formula gives $\cos\theta = 12/21 = 4/7$, so $\theta \approx 55.2°$, which lies inside the interval.
- As $a \to 1$, the points A and B merge and $\theta \to 0$. As $a \to \infty$, A heads off almost along the x-axis and B approaches O almost along the z-axis, so $\theta \to \frac{1}{2}\pi$. These are the two ends of the interval.
- The factorisation of D was checked exactly for thousands of random rational a, b, c, and the angle formula and its range were checked numerically.

Common slips

- Forgetting to say why a, b, c are distinct and non-zero. Those are exactly the facts that make $D \neq 0$.
- Concluding from $\cos\theta < 1$ alone, without excluding $a = \pm 1$. The strict inequality $u > 2$ is what rules out $\theta = 0$.
- Proving only "no other values". The question also asks you to show that every value in the interval is attained, which needs an explicit a.

Beyond STEP

The curve (t^3, t^2, t) is the twisted cubic, a standard example in algebraic geometry. Part (i) is the fact that no four of its points are coplanar, and the determinant you computed is a Vandermonde determinant, which reappears in polynomial interpolation and in linear independence arguments. Strang's *Introduction to Linear Algebra* covers determinants and linear independence clearly.

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